

# Sample Paper

*This sample paper aims to indicate the type of questions and format that you will see in your mid-term exam. The mark distribution, number of questions and difficulty may be different.*

*(The mark distribution in the mid-term exam*

*Section I: 24 marks*

*Section II: 20 marks*

*Section III: 22 marks*

*Section IV: 16 marks*

*Total:80 marks)*

**The Chinese University of Hong Kong**

**Department of Philosophy**

**Mid-term Examination**

**UGED1111 Logic 邏輯**

**Student Name:** \_\_\_\_\_  
**Student ID:** \_\_\_\_\_  
**Time Allowed:** \_\_\_\_ minutes  
**Total Marks:** 50 (total 80 in midterm)

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**INSTRUCTIONS TO CANDIDATES**

- Put all your answers on the answer sheet provided.
  - Answer written on the question papers will not be marked.
  - There are four sections in total.
  - Assessment weighting: 40% of the whole course.
  - You may use the blank area of the question papers to write drafts.
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### Section I. True/False Questions (12 marks)

Put “T” for true statements and “F” for false statements inside the bracket in response to each question. (2 marks for each; 2 mark will be deducted for wrong answer)

1. Something which is technologically possible must be empirically possible.
2. All premises of a sound argument are true.
3. An argument can be 70% valid.
4. From the Aristotelian standpoint, if “No cats are cute animals” is true, then “Some cats are not cute animals” may not be true.
5. If an argument has all true premises and a true conclusion, then it is a valid argument.
6. “ $P \equiv \sim M$ ” is a well-formed formula.

### Section II Categorical Logic (12 marks)

A. Use Venn diagrams to determine whether each is valid or invalid. (4 marks)

(i)

1. All A are B.
2. All B are C.
3. All A are C.

B. Translate the following arguments into standard form categorical syllogisms and use Venn diagrams to determine whether each is valid or invalid. (8 marks)

- (i) If it is a dream about dogs, it is undesirable. Only nightmares are undesirable.  
Therefore, every nightmare is a dream about dogs.

### Section III. Complete Truth Table (16 marks)

(i) Construct a complete truth table for “ $P \vee (\sim P \vee Q)$ ” (4 marks)

(ii) What is/are the logical property of the proposition “ $P \vee (\sim P \vee Q)$ ” (2 mark)

(i) Construct a complete truth table for two compound propositions. (8 marks ; 0.5 mark will be deducted for any wrong or blank entry) and

$P \cdot (Q \vee S)$  and  $P \vee (Q \cdot S)$

(ii) What is/are the logical relation(s) of the two propositions? (2 marks)

### Section IV. Indirect Truth Table (10 marks)

Use indirect truth table to check if the following argument is valid or not.

(You need to indicate the order of reasoning)

$(P \vee M) \supset Q / \sim M / P \equiv \sim M // Q$

## ANSWER

Student Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

### Section I. True/False Questions (10 marks)

1. ( )    2. ( )    3. ( )    4. ( )    5. ( )    6. ( )

### Section II Categorical Logic (14 marks)

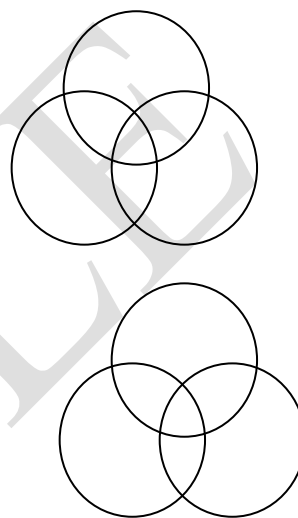
A.

1. All A are B.

2. All B are C.

3. All A are C.

B.



### Section III. Complete Truth Table (16 marks)

1. (i)  $P \vee (\sim P \vee Q)$

| P | Q | P | $\vee$ | ( $\sim$ | P | $\vee$ | Q) |  |  |  |  |
|---|---|---|--------|----------|---|--------|----|--|--|--|--|
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(ii) What is/are the logical property of the proposition? (2 mark)

\_\_\_\_\_

2. (i)  $P \cdot (Q \vee S)$  and  $P \vee (Q \cdot S)$

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

(ii) What is/are the logical relation(s) of the two propositions? (2 marks)

Section IV. Indirect Truth Table (10 marks)

1.  $(P \vee M) \supset Q / \sim M / P \equiv \sim M // Q$

[illegible]

The argument is \_\_\_\_\_.

## ANSWER

Student Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

### Section I. True/False Questions (12marks)

1. ( **T** )    2. ( **T** )    3. ( **F** )    4. ( **F** )    5. ( **F** )    6. ( **T** )

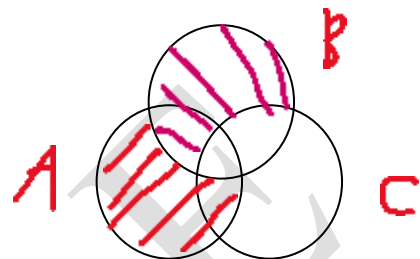
### Section II Categorical Logic (12marks)

1. All A are B.

2. All B are C.

3. All A are C.

A. **Valid**



B.

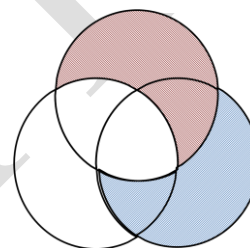
1. All dreams about dogs are undesirable dreams.

2. All undesirable dreams are nightmares.

3. All nightmares are dreams about dogs.

**Invalid**

Nightmares



Dreams  
about dogs

C. All ducks are animals that can swim.

### Section III. Complete Truth Table (16 marks)

1. (i)  $P \vee (\sim P \vee Q)$

| P | Q | P | ~ | P | ~ | Q |   |  |  |  |
|---|---|---|---|---|---|---|---|--|--|--|
| T | T | T | T | F | T | T | T |  |  |  |
| T | F | T | T | F | T | F | F |  |  |  |
| F | T | F | T | T | F | T | T |  |  |  |
| F | F | F | T | T | F | T | F |  |  |  |
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(ii) What is/are the logical property of the proposition? (2 mark)

Tautologous

2. (i)  $P \cdot (Q \vee S)$  and  $P \vee (Q \cdot S)$

| P | Q | S | P | · | (Q | ∨ | S) | / | P | ∨ | (Q | · | S) |  |  |
|---|---|---|---|---|----|---|----|---|---|---|----|---|----|--|--|
| T | T | T | T | T | T  | T | T  |   | T | T | T  | T | T  |  |  |
| T | T | F | T | T | T  | T | F  |   | T | T | T  | F | F  |  |  |
| T | F | T | T | T | F  | T | T  |   | T | T | F  | F | T  |  |  |
| T | F | F | T | F | F  | F | F  |   | T | T | F  | F | F  |  |  |
| F | T | T | F | F | T  | T | T  |   | F | T | T  | T | T  |  |  |
| F | T | F | F | F | T  | T | F  |   | F | F | T  | F | F  |  |  |
| F | F | T | F | F | F  | T | T  |   | F | F | F  | F | T  |  |  |
| F | F | F | F | F | F  | F | F  |   | F | F | F  | F | F  |  |  |
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(ii) What is/are the logical relation(s) of the two propositions? (2 marks)

Consistent

Section IV. Indirect Truth Table (10 marks)

1.  $(P \vee M) \supset Q / \sim M / P \equiv \sim M // Q$

| $(P \vee M) \supset Q$ |   |   |   |   | $/ \sim M /$ |   |  | $P \equiv \sim M$ |   |   | $// Q$ |  |   |  |
|------------------------|---|---|---|---|--------------|---|--|-------------------|---|---|--------|--|---|--|
| T                      | T | F | T | F | T            | F |  | T                 | T | T | F      |  | F |  |
| ⑦                      | ⑧ | ③ | ① | ⑤ | ①            | ② |  | ⑥                 | ① | ④ | ③      |  | ① |  |
|                        |   |   |   |   |              |   |  |                   |   |   |        |  |   |  |
|                        |   |   |   |   |              |   |  |                   |   |   |        |  |   |  |
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|                        |   |   |   |   |              |   |  |                   |   |   |        |  |   |  |

The argument is valid.